SIMATS ENGINEERING

SIMATS Online Programming Lab

C PROGRAMMING

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QUESTIONS 25 / 25 completed

Sum Left Diag.

SUM LEFT DIAG.

Write a C program to find the sum of the elements in the left diagonal of a matrix.

PROGRAM EDITOR

C | Run | Save

```c
#include <stdio.h>

int main() {
    int n, i, j, sum = 0;
    int matrix[10][10];  // supports up to 10x10 matrix

    printf("Enter the size of the matrix (n x n): ");
    scanf("%d", &n);

    printf("Enter the elements of the matrix:\n");
    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    // Left diagonal (main diagonal) → elements where i == j
    for (i = 0; i < n; i++) {
        sum += matrix[i][i];
    }

```

OUTPUT

```
Enter the size of the matrix (n x n): Enter the elements of the matrix:

Sum of the left diagonal elements = 15
```

INPUT

```
3
1 2 3
4 5 6
7 8 9
```

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